

Yang-Jun Joo, Ph.D.

Systems & Safety Evaluation Engineer | Conflict Behavior & Systems Evaluation

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SUMMARY

- Systems and safety evaluation engineer with 7+ years building event- and aggregate-level conflict analytics.
- Specialized in open-loop safety evaluation, scalable data pipelines, and triage workflows.
- PI on two externally funded AV safety projects; delivered evaluation methods to partner agencies (Florida DOT) for intervention prioritization.

SKILLS

Programming	Python, SQL, Git, Linux, ETL pipelines, Dashboards & data visualization
ML/Automation	Workflow automation, VLM-based roadway feature extraction
Data/Methods	Large-scale processing (20GB/day of event logs), Physics-based modeling, reliability analysis, uncertainty quantification, Bayesian models, GLMs
Domain	Conflict behavior evaluation, traffic safety, safety evaluation, open-loop testing

EXPERIENCE

University of Central Florida (UCF) - Postdoctoral Scholar Mar 2024 - Present

- Built Python/SQL pipeline that extracts rear-end conflict events from high-resolution logs across 1,000+ intersections; updated screening/triage logic based on findings.
- Developed micro-hotspot identification for signalized intersections to prioritize conflict zones.
- Developed automated triage pipelines for infrastructure features using Vision-Language Models (VLMs) and Google Street View, reducing manual labeling overhead for triage pipelines.
- PI for an externally funded study quantifying passenger risk perception from dashcam near-miss/crash clips; delivered evaluation insights for safety prioritization.

Seoul National University (SNU) - Postdoctoral Researcher Mar 2023 - Feb 2024

- Cataloged scenarios where AVs require infrastructure support and converted them into an evaluation-ready benchmark matrix (scenario taxonomy + pass/fail criteria) for scalable behavior review.
- Developed physics-based conflict metrics and integrated them into analysis pipelines to support triage and evaluation of infrastructure-aided AV behaviors.

Seoul National University (SNU) - Graduate Research Assistant Mar 2018 - Feb 2023

- Performed reliability-based assessment of lane-change risk using observational trajectory data.
- Designed probabilistic crash risk models (Bayesian network) using traffic violation and crash records to estimate individual driver risk.

EDUCATION

Ph.D., Civil & Environmental Engineering, Seoul National University, 2023.

B.S., Civil & Environmental Engineering, Seoul National University, 2018.

SELECTED IMPACT EVIDENCE

- *IEEE Transactions on Intelligent Transportation Systems* (2025): Evaluating the Effect of Signal Adjustment on Rear-End Conflicts Using High-Resolution Event-Based Data.
- *Accident Analysis & Prevention* (2025): Micro-level hotspot identification at intersections using traffic conflict analysis.
- *Analytic Methods in Accident Research* (2023): A generalized driving risk assessment on high-speed highways using field theory.
- Best Paper Award (TRB Urban Transportation Data & Information Systems, 2023; KOTTA, 2021).